BUSN 5000 Project

Exploring Pay Differences between Women and Men

Nicholas Folino

Fall 2024
(updated 25 Nov 24)

Academic honesty statement

I have been academically honest in all of my work and will not tolerate academic dishonesty of others, consistent with UGA's Academic Honesty Policy.

Sign the academic honesty statement by typing your name on the **Signature** line.

Signature: Nicholas Folino

We will not accept submissions that omit a signed Academic Honesty statement.

Introduction

Overview

Over the course of these slides we will be exploring the gender pay gap. Using the March 2022 CPS (Current Population Survey) Annual Social and Economic Supplement we are able to explore this concept. Conclusive results found through this research project indicate a significant gap between wages of males and females.

Data

March 2022 CPS

Standard CPS coverage

The March CPS surveyed around 70,000 housing units. The standard CPS, conducted monthly for over 50 years, interviews 54,000 households, selected as representative households of the country, states, or certain areas of the country, every month for 4 months in order to garner an understanding of changes on a month-to-month or year-to-year basis.

ASEC supplement information

The survey collects information on employment and demographic status of the population such as age, sex, race, marriage status, education, and family, as well as information regarding work experience, income, and other work- related demographic information.

March 2022 CPS Extract

```
cpsmar_e <- read_csv(here("data", "cpsmar_e.csv"))
```

Creating the extract

In the (`cpsmar_e.`) R script an extract called `cpsmar_e.csv` was created using the data from the 2022 March survey from the household file (`hhpub22.csv`) and person file (`pppub22.csv`). In the extract, the variables Age, Gender, Hispanic Orgin, Education Level, Earnings, Hours Worked, Weeks Worked Per Year, Region, County, City, Occupation, Union Member, indicator for children under 6 years old, Smployment Status, House Sequence, and Union Coverage.

Extract sample information

The dataset was restricted to full-time workers older than 16 using the `mutate` script. The dataset contained 52097 observations and 20 variables.

Analysis sample

```
cpsmar_a <- cpsmar_e %>%
  filter(
    age >= 23,
    age <= 62,
    earnings > 0
  ) %>%
  mutate(
    gender = ifelse(female == 1, "Female", "Male"),
    wage = earnings/(hours*weeks),
    lwage = log(wage),
    Black = case_when(race==2~1, TRUE ~ 0),
    south = case_when(region==3~1, TRUE ~ 0),
    married = case_when((marital==1 | marital==2 | marital==3)~1, TRUE ~ 0),
    age_centered = age - 23
  )
```

Analyzing the `cpsmar_a` sample, it contains 46194observations. Filter and mutate was used to restrict the dataset to 23-62 year olds with positive earnings.

Baseline earnings distributions

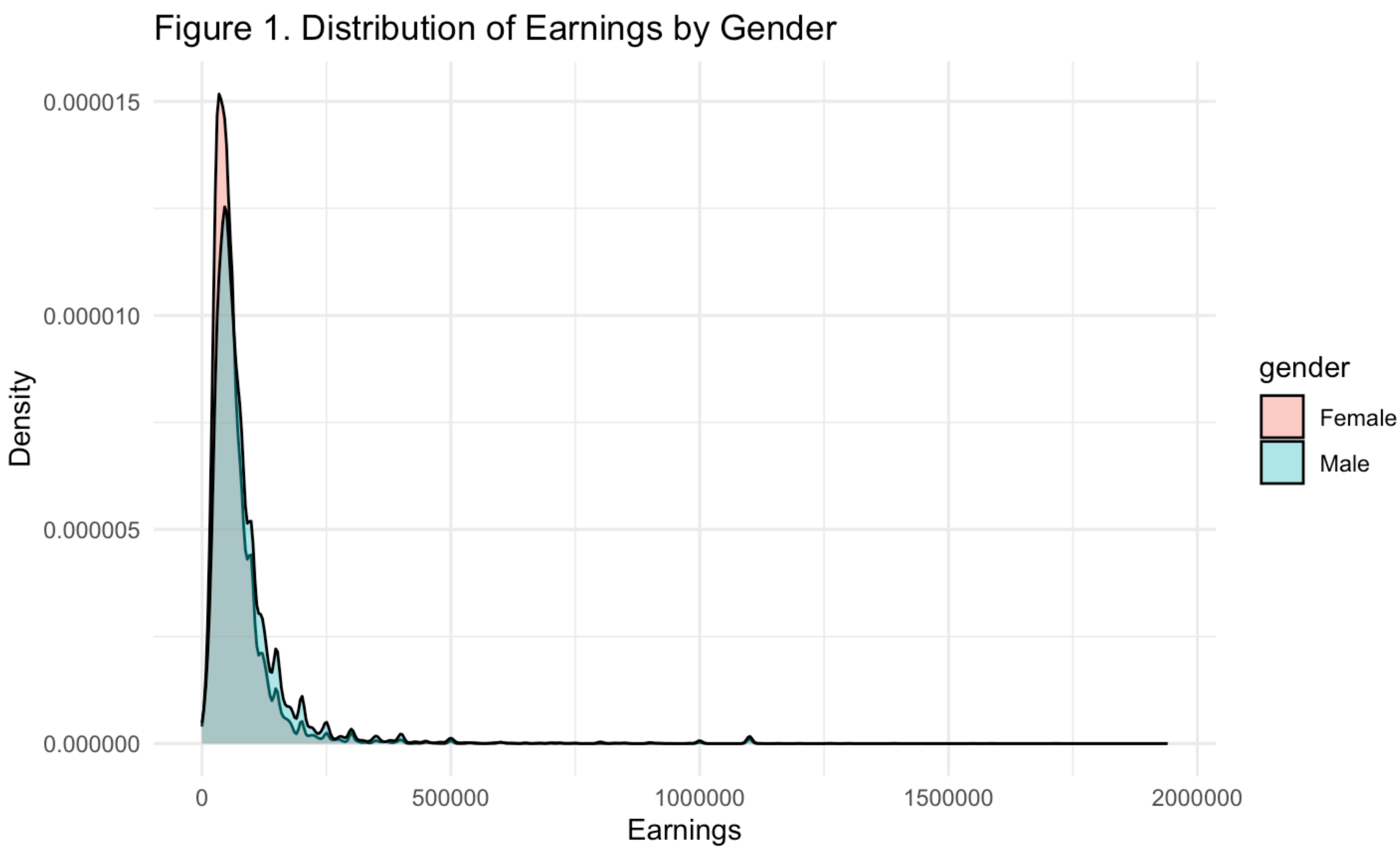
Plotting earnings distributions

```
figure1 <- ggplot(cpsmar_a, aes(x = earnings, group = gender, fill = gender)) +
  geom_density(alpha = 0.4) +
  labs(
    title="Figure 1. Distribution of Earnings by Gender",
    x="Earnings",
    y="Density"
    )+
  theme_minimal()
earnings_fvm <- cpsmar_a %>%
  group_by(gender) %>%
  summarize(avg_earnings = round(mean(earnings, na.rm = TRUE),0))

avg_earnings_f <- earnings_fvm %>%
  filter(gender == "Female") %>%
  pull(avg_earnings) # `pull` extracts the "avg_earnings" value for "Female" from earnings_fvm

avg_earnings_m <- earnings_fvm %>%
  filter(gender == "Male") %>%
  pull(avg_earnings) # `pull` extracts the "avg_earnings" value for "Male" from earnings_fvm,
```

Distribution of earnings by gender



Baseline comparisons

Figure 1 description

Figure 1, titled “Distribution of Earnings by Gender” shows the density distribution for both men and women’s earnings.

Earnings information

The average earnings for men is about $86881 and for females is about $67646. The difference between the men and women average earnings is $19235 dollars, or about $6%.

The career gender gap

Wages and hours differences

Table 1. Wages and hours by gender

	Female			Male		
	N	Mean	SD	N	Mean	SD
earnings	20030	67645.89	71615.67	26164	86880.62	97742.32
wage	20030	30.65	30.92	26164	37.91	41.07
hours	20030	42.26	6.10	26164	44.03	7.77

Documenting the differences

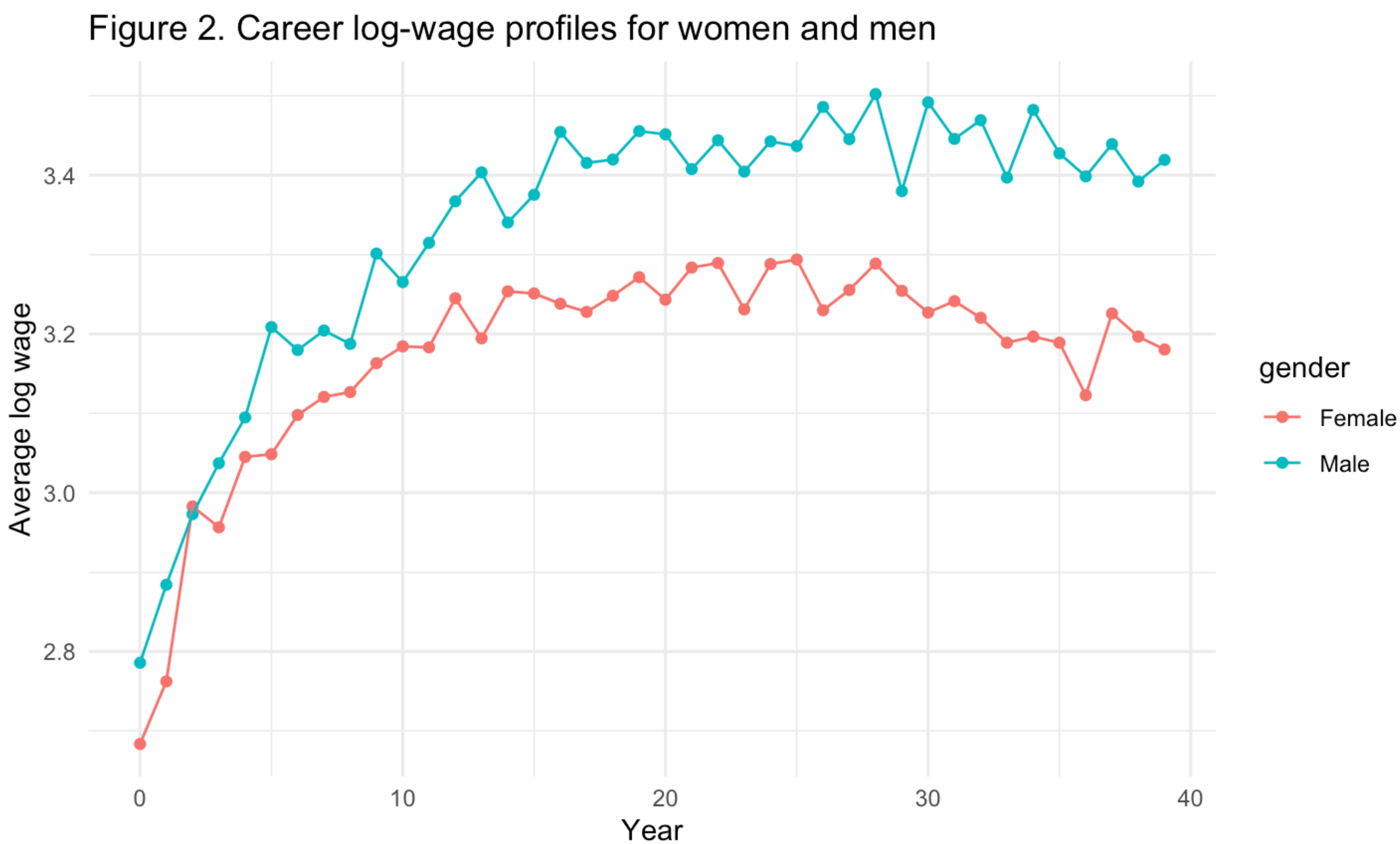
Table 1 demonstrates the disparity between females and male’s, aged 23-62, wages and hours. Males make `7.26` more in wages every hour and also work `1.77`more hours every week.

Plotting career log wage profiles

```
cef_fvm_w <- cpsmar_a %>%
  group_by(gender, age_centered) %>%
  summarize(avg_lwage = mean(lwage, na.rm = TRUE))

figure2 <- ggplot(cef_fvm_w, aes(x = age_centered, y = avg_lwage, color = gender)) +
  geom_point() +
  geom_line() +
  labs(
    title="Figure 2. Career log-wage profiles for women and men",
    x="Year",
    y="Average log wage"
    )+
  theme_minimal()
```

Career log wage profiles



Estimating wage differences over a career

```
males <- cef_fvm_w %>%
  filter(gender == "Male") %>%
  rename(avg_lwage_male = avg_lwage) %>%
  select(-gender)
females <- cef_fvm_w %>%
  filter(gender == "Female") %>%
  rename(avg_lwage_female = avg_lwage) %>%
  select(-gender)

diff_fvm <- inner_join(males, females, by = "age_centered") %>%
  filter(age_centered <= 30) %>%
  mutate(
    diff = avg_lwage_male - avg_lwage_female,
    age_group = cut(
      age_centered,
      breaks = c(-1, 10, 20, 30),
      labels = c("1-10", "11-20", "21-30"))
    ) %>%
  group_by(age_group) %>%
  summarize(mean_diff = mean(diff)*100)

table2 <- kable(
  diff_fvm,
  digits = 2,
  col_names = c("Year Range", "Avg Pct Difference")
```

Evolution of the gender wage gap

Table 2. Percent wage differences, first 30 years

Year Range	Avg Pct Difference
1-10	8.64
11-20	16.42
21-30	18.00

Discussing the gender wage gap evolution

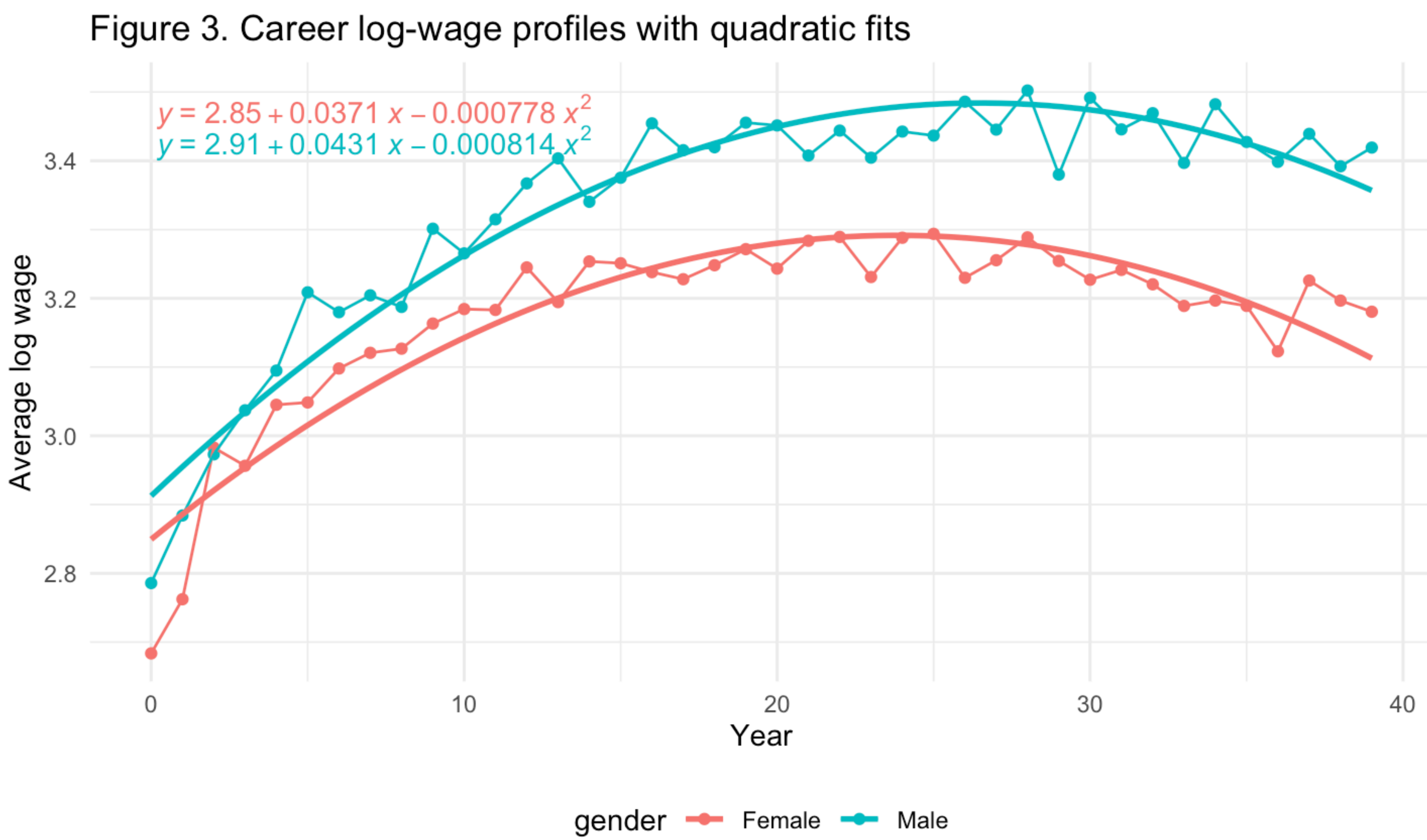
Males and females, at the start of their careers have a smaller gap, as seen in Figure 2, of just 8.64% in the first 10 years. But by 21 -30 years this gap more than doubles to 18% with men making signifigantly more, as seen in Table 2.

Explaining the gender wage gap

Fitting the log wage profiles

```
formula <- y ~ x + I(x^2)
figure3 <- figure2 +
  geom_smooth(
    method = "lm",
    formula = formula,
    aes(group = gender),
    se = FALSE
    ) +
  stat_poly_eq(
    aes(label =  after_stat(eq.label)),
    formula = formula,
    parse = TRUE
    ) +
  labs(
    title="Figure 3. Career log-wage profiles with quadratic fits",
    x="Year",
    y="Average log wage"
    )+
  theme_minimal()+
  theme(legend.position = "bottom")
```

Log wage profiles with quadratic fits



Gender differences in education

Table 3. Educational attainment by gender

	Female		Male	
	N	Mean	N	Mean
CollDeg	20030	0.51	26164	0.41
SomeColl	20030	0.25	26164	0.24
HSGrad	20030	0.20	26164	0.28

Gender differences in demographics

Table 4. Demographic characteristics by gender

	Female		Male	
	N	Mean	N	Mean
Black	20030	0.13	26164	0.09
hisp	20030	0.17	26164	0.21
south	20030	0.38	26164	0.37
city	20030	0.68	26164	0.68
married	20030	0.58	26164	0.66
child_u6	20030	0.19	26164	0.23
age	20030	42.36	26164	42.21

Documenting differences in characteristics

Differences in educational attainment

Table 3 demonstrates the differences in education based on gender. As seen in the table, more Females (51%) than Males (41%) have a college degree. Contrarily, Males have a higher percentage (28%) of high school graduation when compared to females (20%).

Differences in demographic characteristics

Table 4 demonstrates that males have a lower likelihood to be Black, but a higher prevalence of being hispanic than females (9% males vs 13% females for Black) and (21% males vs 17$ female for hispainic)

Controlling for education and demographic characteristics

```
singles <- cpsmar_a %>%
  filter(
    married==0,
    child_u6==0
    )
models <- list(
  "Baseline"      = lm(lwage ~ female +
                       age_centered + I(age_centered^2),
                       cpsmar_a),
  "Add Education" = lm(lwage ~ female +
                       age_centered + I(age_centered^2) + SomeColl + CollDeg + HSGrad,
                       cpsmar_a),
  "Add Person"    = lm(lwage ~ female +
                       age_centered + I(age_centered^2) + HSGrad + SomeColl + CollDeg +
                       Black + hisp + south + city,
                       cpsmar_a),
  "Add Household" = lm(lwage ~ female +
                       age_centered + I(age_centered^2) + HSGrad + SomeColl + CollDeg +
                       Black + hisp + south + city +
                       married + child_u6,
                       cpsmar_a),
  "Only Singles"  = lm(lwage ~ female +
```

Reporting the results

```
cm <- c(
  'female'               = 'Female',
  'age_centered'         = 'Age',
  'I(age_centered^2)' = 'Age$^2$',
  '(Intercept)'          = 'Constant'
)
gm <-  tibble::tribble(
  ~raw, ~clean, ~fmt,
  "nobs", "$N$", 0,
  "r.squared", "$R^2$", 2
)
rows <- tribble(~term, ~Baseline, ~Add_Education, ~Add_Person, ~Add_Household, ~Only_Singles,
  'Education controls',   ' ',          'X',             'X',           'X',             'X',
  'Demographic controls', ' ',          ' ',             'X',           'X',             'X',
  'Household controls',   ' ',          ' ',             ' ',           'X',             'X'
)
attr(rows, 'position') <- c(9, 10, 11) # Positions where you want these rows to appear

table5 <- modelsummary(
  models,
  add_rows = rows,
  coef_map = cm,
  gof_map = gm,
  vcov = c("robust","robust","robust","robust","robust"),
  title = "Table 5. OLS estimates of the gender wage gap"
```

Explaining the gender wage gap

Table 5. OLS estimates of the gender wage gap

	Baseline	Add Education	Add Person	Add Household	Only Singles
Female	-0.162	-0.236	-0.227	-0.214	-0.123
	(0.007)	(0.006)	(0.006)	(0.006)	(0.011)
Age	0.039	0.032	0.031	0.025	0.025
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Age^2	-0.001	-0.001	-0.001	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	2.977	2.479	2.520	2.472	2.495
	(0.010)	(0.017)	(0.018)	(0.019)	(0.031)
Education controls		X	X	X	X
Demographic controls			X	X	X
Household controls				X	X

Documenting the findings

In `Table 5` *titled "OLS Estimates of the Gender Wage Gap", the baseline regression produces a female coefficient estimate of -0.162 which suggests females make on average 16.2% less money than males. When education is added, this gap rises about 7 points. Household and Person characteristics being added reduces the gap by about 1 percentage point. By limiting to only singles without children under 6 years old, the gender wage gap is only 12.3% which is slightly more than half of the full sample.*

Conclusion

Summary

This study, based on current population survey data in 2022, demonstrates a gap in wages between males and females that is statistically significant. The data suggest that the correlation that exists between age, education, personal characteristics, household, and the gap between genders exists, but the causation of this gap is undetermined. The data presented that the wage gap exists at every part of careers of men and women, but the gap is exasperated the further into careers you go. The gap begins small, but by years 21-30 the gap is much larger between men and women.

Appendix

Data documentation

```r
# Define the variables and their descriptions
variables <- data.frame(
  Variable = c(
    "age",
    "earnings",
    "hours",
    "race",
    "marital",
    "HSGrad",
    "SomeColl",
    "CollDeg",
    "region",
    "female",
    "hisp",
    "fulltime"
    ),
  Definition = c(
    "years; capped at 85",
    "total annual earnings",
    "number of hours worked per week",
    "respondent’s race (1 = White only, 2 = Black only, 3 = AI only, 4 = Asian only, 5 = Hawai
    "marital status (1 = Married civilian, 2 = Married AF, 3 = Married absent, 4 = Widowed, 5 =
    "= 1, if education=39",
    "= 1, if 40<education<=42",
    "= 1, if education>=43"
```

List of main variables with definitions

This is a list of the main variables used in this project with their definitions.

Variable	Definition
age	years; capped at 85
earnings	total annual earnings
hours	number of hours worked per week
race	respondent's race (1 = White only, 2 = Black only, 3 = AI only, 4 = Asian only, 5 = Hawaiian/Pacific Islander only (HP), 6 = White-Black, 7 = White-AI, 8 = White-Asian, 9 = White-HP, 11 = Black-Asian, 12 = Black-HP, 13 = AI-Asian, 14 = AI-HP, 15 = Asian-HP, 16 = White-Black-AI, 17 = White-Black-Asian, 18 = White-Black-HP, 19 = White-AI-Asian, 20 = White-AI-HP, 21 = White-Asian-HP, 22 = Black-AI-Asian, 23 = White-Black-AI-Asian, 24 = White-AI-Asian-HP, 25 = White-Black-AI-Asian-HP, 25 = Other 3 race comb., 26 = Other 4 or 5 race comb.)
marital	marital status (1 = Married civilian, 2 = Married AF, 3 = Married absent, 4 = Widowed, 5 = Divorced, 6 = Separated, 7 = Never married)
HSGrad	= 1, if education=39
SomeColl	= 1, if 40<education<=42
CollDeg	= 1, if education>=43